

Nathan H. Carter

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EDUCATION

- **B.S. Mathematics**, minor Data Analytics, James Madison University 2024–2027
- **A.A.S. General Studies**, Mountain Empire Community College 2022–2024
(*Earned concurrently with High School Diploma via Dual Enrollment*)

PROFESSIONAL EMPLOYMENT

- **Optimization Algorithm and Theory Lab** 2026
Department of Mathematics and Statistics, James Madison University, Harrisonburg, VA
- **DevOps Engineer Intern**, CGI, Lebanon VA 2025–2026
 - Managed lifecycle stability for four major repository forks within the DOTA (Tools and Automation) department; utilized Jenkins and Git to maintain high-integrity CI/CD pipelines for large-scale development branches.
- **Software Engineer Intern**, CGI, Lebanon VA 2025
 - Engineered a Python-based automation framework for code analytics, achieving a 300x increase in processing speed and streamlining cross-departmental data reporting.
- **Undergraduate Researcher** 2025
Department of Education, James Madison University, Harrisonburg, VA
 - Developed a strong foundation in qualitative research design, ethics, and academic writing, complementing a primary focus on quantitative modeling.

AWARDS AND HONORS

- **Jeffrey E. Tickle Scholarship in Mathematics**, James Madison University 2026
- **Phi Kappa Phi Honor Society**, National (Top 7.5% of Junior Class) 2026
- **Centennial Scholars Program**, Full-ride merit-based scholarship 2024–2027
- **President's List**, James Madison University (3.95 GPA) 2025–2026
- **FYRE Research Fellowship**, Selected for First-Year Research Experience 2025
- **Summa Cum Laude**, Mountain Empire Community College (4.0 GPA) 2024

RESEARCH INTERESTS

Statistical Ecology, Bayesian Search Theory, Mathematical Optimization, Education, Mixture Models

PAPERS (PUBLISHED AND IN PROGRESS)

- **In Progress:** A continuation of Using scale mixtures of normals to model continuously compounded returns, Nathan Carter & Dr. Hasan Hamdan

RESEARCH PRESENTATIONS AND POSTERS

- **James Madison University Student Colloquium Talk**, Harrisonburg, Virginia, USA. April 27, 2026
 - Normal Mixture Density Estimation for Major US Stocks: A Four-Year Empirical Study, Undergraduate Talk
- **M.A.A. MD-DC-VA Section Meeting**, Frederick, Maryland, USA. April 24-25, 2026
 - Normal Mixture Density Estimation for Major US Stocks: A Four-Year Empirical Study, Professional Talk
- **James Madison University FYRE Symposium**, Harrisonburg, Virginia, USA. April 27, 2026
 - A Comprehensive Examination of University-Based Induction Support for Early Career Teachers, Undergraduate Talk

SELECTED COURSEWORK

Mathematical Theory	Grade	Statistics & Data Science	Grade
Advanced Calculus I (Analysis)	Fall '26	Applied Linear Regression	A
Abstract Algebra I	A	Intro to Machine Learning	A
Advanced Linear Algebra	A-	Data Analysis & Visualization	A
Fourier Analysis & PDE	Fall '26	Research Project: Mixture Modeling	A
Calculus I, II, & III	A	Introduction to Data	A
Differential Equations	A		
Topology	Fall '26		
Modern Optimization	Fall '26		

* Fall 2026 courses reflect current or upcoming enrollment.

TECHNICAL

- **Coding Languages:** Python, R, Java
- **Markup Languages:** $\text{\LaTeX}$, Beamer, HTML, Microsoft Office
- **Tools:** Mac/Windows/Linux, Git, GitHub, Bash, Vim

TEACHING EXPERIENCE

- **Learning Assistant**, MATH 205: Calculus I, James Madison University Fall 2026
- **Mathematics Tutor**, World Tutor (Grades 1–9) 2024–2025

PROFESSIONAL ORGANIZATIONS

- **Mathematical Association of America (MAA)** 2026–Present
- **American Statistical Association (ASA)** 2026–Present
- **The Honor Society of Phi Kappa Phi** 2026–Present